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System and method for measuring edema at home and measurement pattern for use therewith

Abstract

A system and method for measuring edema, at home, is provided, along with a measuring pattern for use therewith. The system includes a computing system that has an imaging device, a processor, a memory, and a wireless transmitter. The computing system is configured to capture edema data of a patient and send the data to a remote doctor. The pattern can be incorporated into a flexible sheet substrate that can be placed over a portion of a patient's skin. By pressing the flexible sheet substrate into the patient, a depression is formed and the pattern is distorted. The imaging device captures an image of the depression and the distorted pattern. The computing system processes the image and determines a depth of the depression based on the distortion of the pattern.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority from U.S. Provisional Patent Application No. 63/151,952 filed Feb. 22, 2021, which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates to measuring edema. The present invention also relates to remote monitoring of edema.

BACKGROUND OF THE INVENTION

(2) Peripheral edema is a phenomenon in which there is an abnormal infiltration and excess accumulation of serous fluid in connective tissue and/or in a serous cavity in one or more of the body's extremities, such as feet, hands, ankles, calves, wrists, and arms. Peripheral edema can be benign and can correct itself in certain circumstances. It can also be, however, an indication of a variety of diseases, such as congestive heart failure (CHF), liver disease, kidney disease, lymphedema, hypoalbumenia, and chronic venous insufficiency. For example, in CHF, the presence of edema in the extremities (e.g., in the lower extremities) can be a valuable diagnostic marker for the presence of disease. In addition, the progression of the edemic state can be monitored over time and related to the progression of the disease.

(3) The most widely used clinical method for assessing edema is digital manipulation, also known

as the “pitting” method. This assessment is accomplished by pressing into the patient's skin near a bony surface and qualitatively evaluating the degree of pitting. Pitting is the indentation in the swollen tissue that remains following removal of the pressure from the edematous area. Due to the altered tissue composition resulting from edema, there can be a putty-like consistency to the tissue, and the tissue can remain in the indented position for several seconds before returning to its original form. The individual performing the test (doctor or other health care provider) assesses one or more of the depth of the indentation, how much force is required to reach a nearby bone, how long the tissue takes to return to its original state, and skin quality. The level of edema is typically described using a ranking system of one to three, corresponding to slight, moderate, and severe edema.

(4) There is a need for additional methods of measuring edema that are reliable, economical, fast, easy to use, that provide repeatable measurements (from day to day and/or from practitioner to practitioner), and that have the potential to be used at home as well as in doctor's offices, hospitals, clinics, and veterinary settings.

SUMMARY OF THE INVENTION

(5) A feature of the present invention is to provide reliable at home diagnostics for patients with edema.

(6) Another feature of the present invention is to provide edema data of a patient to remote doctors, care partners, clinicians, and other caregivers.

(7) A further feature of the present invention is to provide a pattern, an imaging device, and a computing system that enable remote doctors or other remote caregivers to gather edema data about a patient and that facilitate at home diagnostics of the patient.

(8) To achieve these and other advantages, and in accordance with the purposes of the present invention, as embodied and broadly described herein, the present invention includes a system for measuring and monitoring edema. The system can include a flexible sheet substrate having a pattern. The flexible sheet can be configured to be placed on a surface of skin of a patient. The system can further include an imaging device and a computing system. The imaging device is configured to capture image data including an image of the pattern while the flexible sheet substrate is on the surface of the skin of the patient. The computing system includes a processor and a memory. The memory stores computer-readable instructions that, upon execution by the processor, configure the computing system to process image data captured by the imaging device. The image data includes at least an image of the pattern at a location on the surface of the skin of the patient, and a depression into the patient at the location. Processing the image data includes measuring a distortion of the pattern caused by the depression to determine a depth of the depression into the patient. The system can include a remote computing system. The computing system can further include a wireless transmitter configured to wirelessly transmit processed and/or unprocessed image data to the remote computing system. The remote computing system can be at a doctor's office.

(9) According to one or more embodiments, an edema measuring sheet is provided. The edema measuring sheet can include a top flexible layer of a stretchable material, the top flexible layer having an upper surface and a lower surface and including a pattern that is visible when viewing the upper surface. An adhesive layer can be adhered to the lower surface of the top flexible layer and can be configured to be removably adhered to the skin of a patient. The pattern includes a series of parallel lines, for example, at least four parallel lines, and a grid comprising a plurality of markers that surrounds the series of parallel lines. When the edema measuring sheet is adhered to a location on the skin of the patient, and the location on the skin is depressed to form a depression into the patient, the edema measuring sheet is configured to stretch into and contour with the depression so that the pattern is distorted.

(10) The present invention also relates to methods for measuring and monitoring edema. The measuring method can include the steps of: placing a pattern against a surface of skin of a patient,

at a location; depressing a portion of the skin of the patient at the location of the pattern to cause a depression into the patient; and measuring a distortion of the pattern caused by the depression into the patient to determine a depth of the depression into the patient. The pattern can be part of a sheet of stretchable material. The placing of the pattern against the surface of the skin can comprise adhering a lower surface of the sheet to the surface of the skin at the location, such that the pattern is visible by viewing an upper surface of the sheet. The method can further include a step of capturing image data, with an imaging device. The image data can include an image of the pattern against the surface of the skin of the patient, and the depression at the location. The step of measuring the distortion of the pattern can include processing the image data. The method can further include wirelessly transmitting the image data, a grade or value based on the image data, or both, from the imaging device to a remote computing system, for example, over a wireless network. The remote computing system can be at a clinic, at a doctor's office, or the like.

(11) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and intended to provide a further explanation of the present invention, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention may be more fully understood with reference to the accompanying drawings. The drawings are intended to illustrate, not limit, the present teachings, according to an embodiment of the present invention.

(2) FIG. 1A is a schematic view of a flexible sheet substrate applied to a patient, an imaging device, a computing system, and a remote computing system, according to an embodiment of the present invention.

(3) FIG. 1B is a block diagram of components of the computing system shown in FIG. 1A.

(4) FIG. 2 is a cross-sectional view of a flexible sheet substrate, according to an embodiment of the present invention.

(5) FIG. 3 is a partial cross-sectional view of a flexible sheet substrate applied to a patient, and a side view of a computing system including an imaging device positioned to image the flexible sheet, according to an embodiment of the present invention.

(6) FIG. 4 is a side view of a container and a plurality of flexible sheet substrates disposed in the container, according to an embodiment of the present invention.

(7) FIG. 5 is a schematic view illustrating each of a plurality of steps of a method, according to an embodiment of the present invention.

(8) FIG. 6 is a cross-sectional view of a patient's ankle having a depression formed therein, and a side view of a combined pattern-casting and imaging device casting a pattern on the ankle and imaging the pattern.

(9) FIG. 7 is a partially schematic, perspective view illustrating triboelectric charging upon peeling a non-adhesive pattern sheet from a stack, where each non-adhesive pattern sheet has front and back surfaces of first and second respective materials having disparate triboelectric properties.

(10) FIG. 8 is a perspective view of a dispenser for non-adhesive pattern sheets, having a charge-imparting member mounted near an orifice through which individual non-adhesive pattern sheets can be drawn and charged, according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(11) According to various embodiments, the present invention provides a system for measuring and monitoring edema. The system can comprise a flexible sheet substrate, an imaging device, and a computing system. The flexible sheet substrate can include a pattern and can be configured to be placed on the surface of the skin of a patient. The imaging device can be configured to capture

image data including an image of the pattern while the flexible sheet substrate is on the surface of the skin of the patient. The computing system can comprise a processor and a memory. The memory can store a set of computer-readable instructions that, upon execution by the processor, can configure the computing system to process image data captured by the imaging device. The image data can comprise at least an image of the pattern at a location on the surface of the skin of the patient. The image data can also comprise a depression formed into the patient at the location. The depression can be formed by pressing a finger or probe against the skin of the patient, at the location, to form a depression, recess, indent, or other temporary abnormality in the skin.

Processing the image data can comprise measuring a distortion of the pattern, caused by the depression, to determine a depth of the depression into the patient.

(12) The flexible sheet substrate can comprise a stretchable material. The flexible sheet can have an upper surface and a lower surface, and the pattern can be visible from the upper surface. The lower surface can be configured to be adhered to the skin of the patient at the location. The flexible sheet substrate can comprise, for example, a top flexible layer comprising the upper surface of the flexible sheet. The top flexible layer can comprise a bottom surface. The flexible sheet substrate can comprise an adhesive layer having a first surface adhered to the bottom surface of the top flexible layer, and a second surface opposite the first surface, for example, an adhesive surface opposite the first surface. A removeable liner can be provided that is configured to be adhered to and protect the adhesive surface of the adhesive layer. The removable liner can be provided adhered to the adhesive surface so as to protect the adhesive surface. The removable liner can be peeled away from the adhesive surface, for example, just before use, such that the adhesive surface is exposed.

(13) The pattern on the flexible sheet substrate can comprise a series of parallel lines. The flexible sheet substrate can further comprise a plurality of markers surrounding the pattern, for example, in addition to the pattern and at corners of the pattern. The image data can further comprise an image of the plurality of markers. Upon execution of the computer-readable instructions by the processor, the computing system can be configured to determine an angle of the imaging device relative to the location on the surface of the skin, for example, by using the plurality of markers, relative positioning, angles, triangulation, and the like. The flexible sheet substrate can be of any shape. The flexible sheet substrate can comprise a plurality of corners and each of the plurality of markers can be disposed at a respective corner of the flexible sheet substrate.

(14) The system can further comprise a remote computing system. The computing system can further comprise a wireless transmitter configured to wirelessly transmit image data to the remote computing system. The remote computing system can be located, for example, at a clinic, at a dialysis clinic, at a doctor's office, at a nurse's office, at a hospital, at a caregiver's location, or the like.

(15) The system can further comprise a smart device. The smart device can comprise the imaging device and the computing system. The smart device can be, for example, a smart phone, a tablet, a smart watch, smart glasses, or the like. The smart device can be a smart phone that can comprise, for example, a touchscreen interface. The computer-readable instructions can be in the form of application software loaded on the memory, for example, an app loaded on a smart phone.

(16) According to various embodiments, an edema measuring sheet is provided. The edema measuring sheet can comprise, for example, a top flexible layer and an adhesive layer, and can include a pattern as described herein. The top flexible layer can comprise, consist essentially of, or consist of, a stretchable material. The top flexible layer can have an upper surface and a lower surface and can include the pattern. The pattern can be visible when viewing the upper surface. In other words, if the imaging device focuses on the upper surface, the pattern can be imaged by the imaging device. The adhesive layer can be adhered to the lower surface of the top flexible layer. The adhesive layer can be configured to be removably adhered to the skin of a patient, for example, to be securely adhered to the skin but not to cause tearing, other injury, or a rash when removed.

from the skin. The pattern can comprise a series of parallel lines, for example, at least three, at least four, or more parallel lines. The pattern can include a grid, for example, comprising a plurality of markers that surrounds the series of parallel lines. The edema measuring sheet can be configured such that, when the edema measuring sheet is adhered to a location on the skin of a patient, and the location on the skin is depressed to form a depression into the patient, the edema measuring sheet can stretch into and contour with the depression so that the pattern is distorted. The amount or type of distortion can be analyzed to determine the extent of edema affecting the patient and to determine an appropriate course of action for treating the edema. The edema measuring sheet can further comprise a removable protective liner adhered to the adhesive layer.

(17) The top flexible layer of the edema measuring sheet can comprise a plurality of markers. The top flexible layer of the edema measuring sheet can comprise a plurality of corners. The plurality of markers can include at least one respective marker disposed in each of the corners of the plurality of corners. The pattern can be printed on the upper surface of the top flexible layer. The pattern can be etched into the upper surface of the top flexible layer. The pattern can be embedded in the top flexible layer.

(18) According to various embodiments, a container containing a plurality of edema measuring sheets is provided. The container can comprise an opening and a lid covering the opening. A plurality of edema measuring sheets as described herein can be disposed in the container. Each of the plurality of edema measuring sheets can include a removable liner protecting an adhesive surface of the respective edema measuring sheet.

(19) A method for measuring edema is also provided according to the present invention. The method can involve placing a pattern against a surface of the skin of a patient, at a location, and depressing a portion of the skin of the patient at the location of the pattern. The depressing can be of sufficient force to cause the formation of a depression into the patient. The method can involve measuring a distortion of the pattern caused by the depression into the patient. The measuring can determine a depth of the depression into the patient. Based on the depth measured, determined, or calculated, an appropriate treatment for the patient can be figured out and prescribed.

(20) The method can involve using a pattern that comprises a series of parallel lines. The pattern can be part of a sheet of stretchable material. The sheet can be configured to distort with and to be contoured to the depression into the patient. The method can involve a step of placing the pattern against the surface of the skin by adhering a lower surface of the sheet of stretchable material to the surface of the skin at the location. For example, the placement can result in the pattern being visible by viewing an upper surface of the sheet. The sheet of stretchable material can further comprise a plurality of markers surrounding the pattern. The sheet of stretchable material can comprise a plurality of corners and at least some of the plurality of markers can be disposed at respective corners of the plurality of corners. The method can involve capturing image data, with an imaging device, including an image of the pattern against the surface of the skin of the patient. The method can involve capturing image data, with an imaging device, including an image of the depression at the location. The step of measuring the distortion of the pattern can comprise processing the image data. The processing of the image data can be carried out by a computing system. The computing system and the imaging device can be parts of a smart device, for example, parts of the same smart device. The smart device can be selected from a smart phone, a tablet, a smart watch, smart glasses, and the like. The smart device can be a smart phone comprising a touchscreen interface.

(21) The method can further comprise transmitting image data from the imaging device to a remote computing system, for example, over a wired or wireless network. A remote computing system can be at a doctor's office and the method can comprise wirelessly transmitting image data from the imaging device to a remote computing system at the doctor's office. In various embodiments, the image data can wirelessly-transmitted as a live stream video, to the remote computing system. By using the system, a smart device, and the method described herein, a patient can provide a doctor with important information that the doctor can use to assess edema in the patient, without the need

for an office visit. The system and method can evaluate image data, provide a grade or score, for example, a one, a two, or a three, corresponding to whether the edema is slight, moderate, or severe. The system and method enable results, recommendations, and questions, to be discussed during a telehealth visit between the doctor and the patient, and over time can be used to monitor edema.

(22) Other methods of placing a pattern on the skin of the patient can also, or instead, be used. For example, placing the pattern against the surface of the skin of the patient can comprise casting a light pattern against the surface of the skin of the patient. Placing the pattern against the surface of the skin of the patient can comprise transferring a temporary tattoo of the pattern against the surface of the skin of the patient. Placing the pattern against the surface of the skin of the patient can comprise forming a permanent tattoo of the pattern against the surface of the skin of the patient.

(23) According to various embodiments, instead of an adhesive pad, a non-adhesive pattern sheet can be provided that can cling to a patient's ankle or other body part, via an electrostatic or similar charge. A method is provided that can comprise: (i) treating the non-adhesive pattern sheet to provide the non-adhesive pattern sheet with a non-neutral surface charge; and (ii) overlaying the non-adhesive pattern sheet on a patient's body part, for example, an ankle, such that the non-adhesive pattern sheet clings to the body part due to the non-neutral surface charge.

(24) According to various embodiments, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, and/or at least 95% of the surface area of one side of the non-adhesive pattern sheet is in intimate clinging contact with the body part. Such surface-area percentages are considered herein to comprise a substantial portion of the non-adhesive pattern sheet. In some embodiments, about 25%, or about 50%, or about 75% of the surface area of one side of the non-adhesive pattern sheet can be caused to cling to the surface of the body part for a period of time (for example, for greater than 20 seconds).

(25) The treating of the non-adhesive pattern sheet can comprise tribocharging (also known as "triboelectric charging," "contact charging" or "electrification by contact"). The treating can include passing the non-adhesive pattern sheet through an electric field.

(26) A charge differential can be established by triboelectric charging, inducing polarization, passage through an electric field, and the like. The charge differential can be at least 5V, at least 7V, at least 10V, at least 15V, at least 20V, at least 25V, at least 30V, at least 50V, at least 75V, at least 100V, or higher.

(27) A plurality of non-adhesive pattern sheets can be provided. The plurality can comprise a laminar (multi-layer) sheet-like material having a first outer layer on one side and a second outer layer on an opposite side. The first and second layers can be comprised of first and second materials, with the first and second materials having unlike triboelectric properties. Accordingly, when surfaces of the first and second materials are brought into intimate contact with one another, and then separated, each acquires a non-neutral surface charge. The plurality of non-adhesive pattern sheets can be releasably connected to one another, end-to-end, and wound so as to define a roll configuration. The plurality of such sheet-like materials can be arranged in a stack, with each sheet-like material of the stack disposed in intimate contact with its nearest neighbors (or neighbor, in the case of the first and last sheets). The first layer of each sheet-like material of the stack can be disposed facing in a first direction and the second layer of each sheet-like material of the stack can be disposed facing in a second direction, wherein the second direction is opposite of the first direction. One of the layers can comprise, for example, of a paper material, for example, tissue, and the other of the layers can optionally be provided with adhesive properties, for example, a slight degree of stickiness or tackiness. At least one of the layers can comprise an electret.

(28) A further aspect of the present invention provides a dispensing device for dispensing non-adhesive pattern sheets. The dispensing device can comprise: a compartment adapted to hold a plurality of non-adhesive pattern sheets; an orifice defined by a wall of the compartment permitting

one of the non-adhesive pattern sheets to be pulled therethrough for removal from the compartment; and a charge-imparting member positioned such that a non-adhesive pattern sheet pulled through the orifice must pass closely by, for example, rub against, the charge-imparting member. In use, a non-adhesive pattern sheet removed from the dispenser via the orifice gains a non-neutral surface charge. The charge-imparting member and the non-adhesive pattern sheets can have unlike triboelectric properties. For example, non-adhesive pattern sheets can be made of a paper material and the charge-imparting member can comprise a vinyl or Styrofoam surface. The charge-imparting member can be an electrostatic generator.

(29) According to various embodiments, the present invention exploits the phenomenon known as “triboelectric charging,” or “tribocharging” for short. This phenomenon is also referred to as “contact charging” or “electrification by contact.” There are several mechanisms that can contribute to the resulting charge that is generated by the triboelectric process. Four major factors that have the greatest influence on the triboelectric charging process are: surface contact effects; work function; charge back flow; and gas breakdown. These factors and other materials, systems, and methods that can be used in accordance with the present invention are described in U.S. Pat. No. 6,484,328 B1 to Frazier, which is incorporated herein in its entirety by reference.

(30) In accordance with various embodiments of the present invention, a non-adhesive pattern sheet is tribocharged before it is finally placed on a patient's body part. The charge established on the non-adhesive pattern sheet provides a “cling” characteristic that assists in maintaining the non-adhesive pattern sheet in place once situated on the body part. The affinity between the non-adhesive pattern sheet and the body part can be sufficient to maintain the non-adhesive pattern sheet in intimate clinging contact with the body part for at least 10 seconds, at least 15 seconds, at least 20 seconds, at least 30 seconds, at least 45 seconds, and/or at least one minute.

(31) The non-adhesive pattern sheet can be incorporated into a square or rectangular sheet. Perforations or scoring can allow non-adhesive pattern sheets cover to be readily torn from the roll of such sheets. The non-adhesive pattern sheet is macroscopically two-dimensional and planar, although not necessarily flat. The non-adhesive pattern sheet has some thickness in the third dimension, but the third dimension is very small compared to the first two dimensions. The non-adhesive pattern sheet can exhibit good drapability. Drapability is the tendency of the non-adhesive pattern sheet to conform, normal to its plane, to a body part, such as a patient's ankle upon which the non-adhesive pattern sheet is placed. The higher the drapability of the non-adhesive pattern sheet, the more accurately it will conform to the body part. More accurate conformation to the body part can provide greater retention thereon. The non-adhesive pattern sheet can comprise a paper material, a soft tissue-type paper, a textile material, a cloth material, or the like. For a variety of reasons, in certain embodiments of the invention, fibrous non-adhesive pattern sheets, for example, comprising wood pulp, cellulosic, or paper-based materials, such as tissue-type paper, can generally be preferred. For example, such materials are most compatible with existing sewage and treatment facilities. Paper-based products are typically better in terms of biodegradability. A coating, for example, a water receptive wax, can be applied to form one or more layers on a paper sheet.

(32) Tribocharging can be caused during unwinding of a non-adhesive pattern sheet from a roll. The non-adhesive pattern sheet can be provided as a roll, in the nature of a roll of paper towels or toilet paper, such that a user partially unwinds the roll to obtain the next-available non-adhesive pattern sheet which is then torn from the roll. Tearing can be aided by a perforation line or scoring. The non-adhesive pattern sheet can be a laminate article comprising upper and lower sides of different materials with a spaced-apart ranking in the triboelectric series. For example, one side of each non-adhesive pattern sheet can be a paper-based tissue-type material, and the other side can be a rubber-based film layer, so that tribocharging occurs upon unrolling in a fashion analogous to the tape-dispenser. A dispenser in the nature of an elongated rod can be utilized to receive a roll in a manner permitting unwinding of the roll thereabout.

(33) So-called “cling films” can be used as the non-adhesive pattern sheet material. Nonlimiting

examples of films include polyvinyl chloride films (whether plasticized or flexible), polyurethane films, high density polyethylene films, polypropylene films, tackified linear low density polyethylene films, acrylonitrile/butadiene/isoprene films, acrylonitrile/butadiene copolymers, styrene/butadiene copolymers (whether random or block), styrene/isoprene copolymers (whether random or block), styrene/ethylene-butylene block copolymers, and combinations thereof, as well as other films known in the art for making cling films. A “cling” property can be imparted to plastic film by including a high content of plasticizer in the plastic. Polyvinyl chloride films with this property are commercially available from several sources. An example of film having a cling property is Sure-State Series 9000, which is available from Tekra Corporation, New Berlin, Wisconsin. Permacharge Corp., Rio Rancho, N. Mex., provides an electrostatic polypropylene material called ClingZ. ClingZ gets its adhesive properties from a permanent electrostatic charge. The film is repositionable like other cling films, but ClingZ contains no plasticizers. The present invention contemplates the use of such material, as well, in the make-up of a non-adhesive pattern sheet herein. The non-adhesive pattern sheet can comprise an electrostatically charged nonwoven fabric.

(34) Referring to FIGS. 1A and 1B, FIG. 1A depicts a system for measuring and monitoring edema **100**, according to an exemplary embodiment of the present invention. FIG. 1B depicts a block diagram of a computing system **120**, according to an exemplary embodiment of the present invention. System **100** can include a flexible sheet substrate **132** having a pattern **104**. System **100** includes an imaging device **116** and computing system **120**. Computing system **120** can comprise imaging device **116**, a processor **122**, a memory **126**, computer readable instructions stored in memory **126**, and a wireless transmitter **130**. A patient **108**, a physician, a technician, a caregiver, a care partner, or another user can place flexible sheet substrate **132** on a surface of the skin **112** of the patient at the patient's foot, hand, ankle, calf, wrist, arm, or the like. The user then presses into the patient's skin **112** at a location overlaid by pattern **104** to cause a depression **128**. Depression **128** can be formed by pressing a finger or probe against skin **112** of patient **108**, at the location. If any sort of depression **128** remains in the patient after the pressure is released, the user captures an image of the depression and pattern, or, more generally, image data, using imaging device **116**. Imaging device **116** can be a cell phone camera and computing system **120** can be a cell phone. The image data can be captured by the imaging device **116** within 1 second, within 2 seconds, within 3 seconds, within 4 seconds, within 5 seconds, within 6 seconds, within 8 seconds, within 10 seconds, within 15 seconds, or longer, after the pressure is released. The image data includes an image **124** of pattern **104** on the surface of skin **112** of patient **108**, and depression **128** into patient **108** at the location of pattern **104**. Computing system **120** then processes the image data by measuring a distortion of pattern **104** caused by depression **128** to determine a depth of depression **128** into patient **108**.

(35) Computing system **120** can be a smart device, laptop, desktop, server, or the like. Examples of smart devices can include a smart phone, a tablet, a smart watch, and smart glasses. As illustrated in FIG. 1A, computing system **120** can include a smart phone that has a touch screen interface **144**. The smart phone can comprise computing system **120**, including imaging device **116**. In such embodiments, the computer-readable instructions are in the form of application software (app) loaded in the memory of the smart phone. After depression **128** is made, the user can capture an image **124** of the depression using imaging device **116** and touch screen interface **124** of the smart phone.

(36) Although pattern **104** is illustrated as being part of flexible sheet substrate **132**, pattern **104** can be provided via a different medium. For example, pattern **104** can be drawn on the patient using a marker, pattern **104** can be projected onto skin **112** by a light projector, pattern **104** can be part of a permanent or temporary tattoo, or the like. Pattern **104** can include a series of parallel lines each disposed in a horizontal direction and located above and below one another. Pattern **104** can include a series of parallel lines each disposed in a vertical direction and located side by side.

Pattern **104** can include a series of parallel lines disposed in diagonal directions. Pattern **104** can include a grid pattern including a plurality of vertical and horizontal lines crisscrossing one another. Pattern **104** can include a series of dots, a series of wavy lines, a series of zigzags, or any pattern that can be preloaded on memory **126** and recognized by computing system **120**. If a series of parallel lines, the pattern can comprise, for example, at least three, at least four, at least five, at least six, or more parallel lines.

(37) Flexible sheet substrate **132** can further include a plurality of markers **136** surrounding pattern **104**, for example, a quick response (QR) code. Plurality of markers **136** can comprise markers in each corner of the flexible sheet substrate, in three of four corners of the flexible sheet substrate, in an arc partially surrounding the pattern, in a circle surrounding the pattern, or the like. Each marker of plurality of markers **136** can independently be any shape, such as a circle shape, a square shape, a rectangular shape, a triangular shape, a trapezoid shape, a star shape, or another shape. As illustrated in FIG. **1A**, markers **136** can include circle shapes disposed at each of the corners of flexible sheet substrate **132**. The plurality of markers **136** allows computing system **120** to perform geometric calculations to determine where, and at what angle, imaging device **116** is situated relative to pattern **104**.

(38) When computing system **120** and imaging device **116** are part of a smart device, the app can guide the user to capture the image. For example, the smart device can include a display. The app can guide the user and prompt the user to capture an image that can be readable by computing system **120** by displaying brackets, a rectangle, a square, lines, dots, or other imaging guides on the display that overlay a live feed from imaging device **116**. The app prompts the user to capture the image when pattern **104** is within the imaging guide during the live feed of the imaging device **116**. This ensures that computing system **120** is capable of processing the captured image to determine the depth of depression **128**. The app can also provide a timer for computing system **120** to measure a period of time. For example, when pressure is released from skin **112**, timer can be activated by the user pressing an activator on touchscreen interface **144**. Timer can automatically stop when the image is captured by the user. An amount of time between when pressure is released and when the image is captured, can be recorded. The amount of time can be superimposed on the image or can be incorporated with the image data in other fashions. The length of time between when pressure is released and when the image is captured, can be used to adjust, calibrate, or modify the image data or the image data and processor can otherwise calculate a score of the depression based, in-part, on the length of time. If the length of time between when pressure is released and when the image is captured, is too long, an error message or signal can be sent or generated or both.

(39) As mentioned above, computing system **120** processes the image data by measuring a distortion of pattern **104** caused by depression **128**, and using the distortion, determines a depth of depression **128** into patient **108**. Computing system **120** can first perform geometric calculations to determine an angle of imaging device **116** relative to skin **112** of patient **108** at the location of pattern **104**, and the geometric calculations can be made by using plurality of markers **136**. A distance between markers **136** can be preloaded on memory **126** of computing system **120**.

(40) Computing system **120** can determine the angle by calculating the difference between the distance of markers **136** in the image compared with the actual distance between markers **136** on the flexible sheet substrate **132** and then calculating the relative angle. Once the angle is determined, computing system **120** can use the angle and the distortion of pattern **104** to determine a depth of depression **128** into patient **108**. Pattern **104** can be preloaded on the memory of computing system **120**. Computing system **120** can compare pattern **104** on the image with a preloaded version of pattern **104** to determine an extent of distortion in pattern **104** and to thereby determine a depth of depression **128**. The depth of depression **128** can be displayed on a display of computing system **120**, a display of a remote computing system **140**, on both, or not at all.

(41) Once the depth of depression **128** is determined, computing system **120** can assign a severity

level of depression **128** and display the severity level to the user, a remote doctor, a caregiver, a care partner, or the like. For example, the levels can be based on a scale such as a three-point scale or a four-point scale. An exemplary four-point scale can include a slightly pitted level 1, a somewhat pitted level 2, a deeply pitted level 3, and a severely deeply pitted level 4. Computing system **120** can thereby determine the severity of depression **128** and can display the severity to the user, remote doctor, caregiver, care partner, or the like. Other types of grading scales can be used and presented on a display.

(42) Computing system **120** can further include a wireless transmitter **130** that wirelessly transmits data over a wireless network **148**. For example, computing system **120** can send data to remote computing system **140** or remote servers over the Internet. Image data can be processed on computing system **120**, at remote computing system **140**, or both. Processed and unprocessed image data, as well as severity levels, can be saved on the memory of computing system **120**, on a memory of remote computing system **140**, on a remote server such as a cloud-based server, or on a combination thereof.

(43) Determining the depth of depression **128** can be performed on computing system **120** that is hard wired to imaging device **116**, on a remote computing system **140**, or both. For example, the user can capture the image data and the depth can be calculated on the app of the smart device. Once the depth of depression **128** is calculated, depth data including the depth of depression **128**, severity level, or both, can be sent to remote computing system **140** over wireless network **148**. In certain embodiments, the user can capture the image data and then the image data is sent to remote computing system **140** over wireless network **148** for remote computing system **140** to determine the depth of the depression **128**, the severity level, or both.

(44) Additionally, image data, depth data, or both can be combined with other diagnostic data to be sent to remote computing system **140**. Other diagnostic devices that capture diagnostic data of the patient can communicate with computing system **120** via BLUETOOTH, WIFI, or other wired or wireless connections, to combine the diagnostic data with the image data, depth data, severity level, or a combination thereof. For example, heart rate, blood pressure, video screenings, audio screenings, or other diagnostic data of the patient can be captured by one or more diagnostic devices and can be sent over a wire or wirelessly to computing system **120** or to a central computing hub. The diagnostic data can be combined with the image data, depth data, severity level, or a combination thereof as a data package and the data package can be sent to remote computing system **140** for a remote doctor, or the like, to review.

(45) Remote computing system **140** can be at a clinic, at a dialysis clinic, at a doctor's office, at a nurse's office, at a hospital, at a caregiver's location, at a veterinary's office, or the like. System **100** of the present invention allows a remote patient to send diagnostic data, image data, depth data, severity level, or a combination thereof to remote computing system **140** of a remote doctor such that the remote doctor can diagnose the patient remotely. Thus, the present invention can be used to prevent exposure of patients to environments conducive to contracting diseases, such as COVID-19, or other viral and bacterial infections. Once the remote doctor has received diagnostic data, image data, depth data, severity level, or a combination thereof, and the depth of depression **128** is determined, the remote doctor can determine an extent of edema of the patient and prescribe appropriate treatments. Additionally, each of the diagnostic data, image data, depth data, severity level, or a combination thereof can be stored on a memory of either computing system **124**, remote computing system **140**, or a remote server. Depths of multiple depressions, captured over time, can be saved and compared. Images over time can be processed to provide a time lapse, or the like, to determine if a patient's health is improving or deteriorating.

(46) FIG. 2 illustrates a cross sectional view an edema measuring sheet that includes flexible sheet substrate **232**. Flexible sheet substrate **232** can include a top flexible layer **206** and a pattern layer or pattern **204**. Top flexible layer **206** can be placed against the skin of a patient such that pattern **204** is covering a portion of the skin. Top flexible layer **206** can be attached to the skin by an

adhesive, by electrostatic forces, by tribocharged forces, by being held by a user, or the like.

(47) Flexible sheet substrate **232** can have a width and a length, each independently of from 1 inch to 10 inches, from 2 inches to 8 inches, from 4 inches to 6 inches, or of 5 inches. Flexible sheet substrate **232** can have a thickness of from 6 μm to 6 mm, from 10 μm to 2 mm, from 20 μm to 1 mm, or from 50 μm to 0.5 mm. Flexible sheet substrate **232** can be of any shape, such as a square shape, a rectangular shape, a circular shape, an oval shape, a trapezoid shape, or the like, and can have a flat profile made of a flexible and stretchable material. For example, flexible sheet substrate **232** can have rectangular shape with four corners. Flexible sheet substrate **232** can be capable of stretching from 10% up to 50%, from 15% up to 45%, from 20% up to 40%, from 25% up to 35%, or 30% beyond its resting length, resting width, or both. Top flexible layer **206** can be made of a stretchable material such as, but not limited to, a flexible fabric, a flexible polymer, or a combination thereof. The flexible polymer can include a thin sheet of polyethylene, polypropylene, ethylene vinyl acetate (EVA), polyurethane, and the like.

(48) Top flexible layer **206** can have an upper surface **218** and a bottom surface **220**. Top flexible layer **206** further includes pattern **204** that is visible when viewing upper surface **218**. As illustrated in FIG. 2, pattern **204** is printed on or attached to upper surface **218** of top flexible layer **206**. Alternatively, pattern **204** can be etched on the upper surface of top flexible layer **206**, embedded inside of top flexible layer **206**, screen printed on top flexible layer **206**, laser printed on top flexible layer **206**, or the like.

(49) Top flexible layer **206** can include any color that contrasts with a color of pattern **204** such that pattern **204** is easily visible. For example, top flexible layer **206** can be white, while pattern **204** can be black, or vice versa. In certain embodiments, top flexible layer **206** can be opaque and pattern **204** can include a dark color that is easily visible when applied to the skin of the patient.

(50) Flexible sheet substrate **232** can further include an adhesive layer **210**. Adhesive layer **210** can include a first surface adhered to bottom surface **220** of top flexible layer **206**. Adhesive layer **210** can also have an adhesive surface opposite the first surface. The adhesive that forms adhesive layer **210** can be sufficiently flexible to follow the skin, can be irritant-free to the skin, can be free of chemicals that would cause chemical stimulation, can be free of properties that would cause physical stimulation, and can have moisture permeability. For example, the adhesive can be selected from synthetic rubber adhesives, styrene-isobutylene-styrene copolymer adhesives, polyurethane adhesives, polysiloxane adhesives, natural rubber adhesives, polyether adhesives, acrylic adhesives, wax adhesives, or a combination thereof. The adhesive can contain various additives, if desired, for example, plasticizers represented by polyvalent alcohols such as glycerol, polyethylene glycol, polypropylene glycol, and the like, aqueous or water-absorbable resins such as polyacrylic acid, crosslinked polyacrylic acid, polyvinylpyrrolidone, and the like, rosin, terpene, petroleum tackifiers, softeners, fillers, pigments, thickeners, and the like. Adhesive layer **210** is removably adhered to the skin of a patient, for example, to be securely adhered to the skin but not to cause tearing, other injury, or a rash when removed from the skin. For example, the adhesive strength can be 10 g/inch to 200 g/inch, 30 g/inch to 150 g/inch, 40 g/inch to 100 g/inch, 50 g/inch to 70 g/inch, or 60 g/inch.

(51) A removable protective liner **214** can adhere to the lower adhesive surface of adhesive layer **210** and can cover adhesive layer **210**. Removeable liner **214** can protect adhesive layer **210** from debris and moisture and can preserve the adhesiveness of adhesive layer **210**. Removeable liner **214** can be peeled away from the lower adhesive surface such that the adhesive surface is exposed so that the edema measuring sheet can be adhered to the skin of the patient when ready for use.

(52) FIG. 3 is a cross-sectional view of a flexible sheet substrate **332** wherein a removeable liner has been peeled away from an adhesive layer **310** and adhesive layer **310** has been adhered to a patient's skin **312**. As illustrated in FIG. 3, a depression **328** is formed in patient **308** and a pattern **304** has been distorted. An imaging device **316** of a computing system **320** captures image data of depression **328** and computing system **320** is configured to determine a depth of depression **328**.

based on the distortion of pattern **304**.

(53) FIG. **4** illustrates a container **450** containing a plurality of edema measuring sheets **432**. Container **450** can include an opening **458** and a lid **454** that can cover the opening or be hinged-back to enable access to edema measuring sheets **432**. Lid **454** can be connected to container **450** by a hinge, a friction fit, clasps, locks, ties, magnets, or the like. A plurality of edema measuring sheets **432**, as described herein, can be disposed in container **450**. Each of the plurality of edema measuring sheets **432** can include a removable liner protecting an adhesive surface of the respective edema measuring sheet **432**. Alternatively, edema measuring sheets **432** can be stored in container **450** in the form of a roll. Lid **454** can be opened, exposing edema measuring sheets **432** such that at least one edema measuring sheet **432** can be removed and used. Edema measuring sheet **432** can be a disposable item or can be reusable and placed back into container **450**. After removing an edema measuring sheet **432**, lid **454** can be closed and container **450** can be stored. Edema measuring sheets **432** can be packaged and shipped in container **450**. Additionally, container **450** can be used for storage of edema measuring sheets **432** at a home of a patient, at a doctor's office, at a clinic, at a hospital, at a veterinary office, or the like.

(54) FIG. **5** illustrates a method for measuring edema, which involves a number of steps **500**. Steps **500** can involve placing a pattern **504** against a surface of a skin **512** of a patient, at a location, and depressing a portion of the skin of the patient at the location of pattern **504**. The depressing can be of sufficient force to cause the formation of a depression **528** into the patient. Steps **500** can involve measuring a distortion of pattern **504**, caused by depression **528** into the patient. The measuring can determine a depth of depression **528** into the patient. Based on the depth measured, determined, or calculated, an appropriate treatment for the patient can be figured out and prescribed. Over time, the edema can be monitored.

(55) Pattern **504** includes a series of parallel lines or another pattern **504** as described herein. Pattern **504** can be part of a sheet **532** of stretchable material, can be a light projection, can be drawn onto skin **512** by a marker, can be part of a temporary or permanent tattoo, or the like. Sheet **532** can be configured to distort with and to be contoured into depression **528** in the patient. Steps **500** can involve a step of placing pattern **504** against the surface of the skin by adhering a lower surface of sheet **532** of stretchable material to the surface of skin **512** at the location. Alternatively, sheet **532** can attach to skin **512** by electrostatic forces or can be held against skin **512** by a user. The placement can result in pattern **504** being visible by viewing an upper surface of sheet **532**. Sheet **532** of stretchable material can further include a plurality of markers **536** surrounding pattern **504**. Sheet **532** of stretchable material can include a plurality of corners and at least some of the plurality of markers **536** can be disposed at respective corners of the plurality of corners. Method **500** can involve capturing image data, with an imaging device, including an image **524** of pattern **504** against the surface of skin **512** of the patient. Steps **500** can involve capturing image data, with an imaging device, including an image **524** of depression **528** at the location. The step of measuring the distortion of pattern **504** can include processing the image data. The processing of the image data can be carried out by a computing system **520**. Computing system **520** and imaging device can be part of a smart device, for example, part of the same smart device. The smart device can be selected from a smart phone, a tablet, a smart watch, smart glasses, and the like. The smart device can be a smart phone having a touchscreen interface **544**.

(56) Steps **500** can further include transmitting image data from the imaging device to a remote computing system **540**, for example, over a wired or wireless network **548**. Remote computing system **540** can be at a doctor's office and steps **500** can include wirelessly transmitting image data from the imaging device to remote computing system **540** at the doctor's office. In various embodiments, the image data can be wirelessly-transmitted as a live stream video, to remote computing system **540**. Method **500** can further include storing image data, depth data, a severity level score, or a combination thereof, to a memory of either computing system **520**, remote computing system **540**, or a remote server. The depths of multiple depressions can be compared.

Images **524** can be further processed by computing system **520** and/or remote computing system **540**. A time-lapse sequence, series, video, or the like, can be generated to determine if a patient's health is improving or deteriorating over time. The method comprising steps **500** enables a patient to provide a doctor with important information that the doctor can use to assess edema in the patient, without the need for an office visit. The method enables results, recommendations, and questions, to be discussed during a telehealth visit between the doctor and the patient.

(57) FIG. **6** is a side view of an exemplary embodiment in which a pattern is projected onto skin **612** of a patient **608** in the form of light, without the need for a flexible sheet. The light pattern can be projected onto the skin by a pattern projector **650** that can project any type of visible light, such as laser light. A computing device **620**, including imaging device **616** and pattern projector **650**, can all be part of the same smart device or parts of devices that are separate from one another. In certain embodiments, pattern projector **650** can be separate from the smart device and removably attached thereto. For example, pattern projector **650** can be part of an adapter that is removably attached to the smart device by clips, adhesives, magnets, ties, mating slots, or the like. Pattern projector **650** can include a fitting and can be configured to fit over a flashlight feature of a cell phone device.

(58) As shown in FIG. **6**, a depression **628** is made into patient **608** by pressing against skin **612**. Pattern projector **650** projects light **654** in the form of the pattern, onto depression **628**. The light pattern is distorted due to depression **628**. Imaging device **616** captures an image of the distorted light pattern and depression **628**. Computing system **620** compares a preloaded pattern to the distorted light pattern that has been captured and calculates the depth of depression **628**.

(59) FIG. **7** is a partially schematic illustration of an embodiment of the present invention wherein a plurality of non-adhesive pattern sheets is arranged in a stack, denoted generally as **730**. Each non-adhesive pattern sheet of the stack is disposed in intimate contact with its nearest neighbors (or neighbor, in the case of the first and last sheets). A biasing mechanism can be provided to keep the non-adhesive pattern sheets in pressing engagement. Each non-adhesive pattern sheet is comprised of at least two layers. The two layers are comprised of materials differing from one another with respect to their triboelectric properties. A first outermost layer of each non-adhesive pattern sheet of the stack is disposed facing in a forward direction and an outermost layer on the opposite side of each non-adhesive pattern sheet is disposed facing in a reverse direction. Triboelectric charging is caused by peeling a non-adhesive pattern sheet **720** from stack **730**. Attention is directed to the plus (+) and minus (-) signs in FIG. **7**, indicating positive and negative surface charges upon peeling. Conveniently, the non-adhesive pattern sheets of the present invention can be provided for use with a dispenser, for example, a dispenser configured to hold, for example, 50 or 100 non-adhesive pattern sheets.

(60) FIG. **8** illustrates a dispenser **828** for non-adhesive pattern sheets, having a charge-imparting member **832** mounted near an orifice **834** through which individual non-adhesive pattern sheets, such as sheets **820**, can be pulled. Charge-imparting member **832** can comprise a vinyl or Styrofoam member mounted at an orifice of dispenser **828**, through which orifice fresh non-adhesive pattern sheets are pulled. For example, a vinyl member can include a surface portion disposed such that a non-adhesive pattern sheet pulled through the orifice is forced to contact it, thereby providing a means for contact electrification. The vinyl member can be bonded to the body of dispenser **828** by means of a suitable insulating adhesive that will not adversely affect its electrostatic properties. An electrostatic generator can be mounted on the dispenser such that it imparts a persistent electrostatic charge to a cover as the cover is removed from the dispenser.

(61) The entire contents of all references cited in this disclosure are incorporated herein in their entireties, by reference. Further, when an amount, concentration, or other value or parameter is given as either a range, preferred range, or a list of upper preferable values and lower preferable values, this is to be understood as specifically disclosing all ranges formed from any pair of any upper range limit or preferred value and any lower range limit or preferred value, regardless of

whether such a range is separately disclosed. Where a range of numerical values is recited herein, unless otherwise stated, the range is intended to include the endpoints thereof, and all integers and fractions within the range. It is not intended that the scope of the invention be limited to the specific values recited when defining a range.

(62) All patents, patent applications, and publications mentioned herein are incorporated herein in their entireties, by reference, unless indicated otherwise.

(63) Other embodiments of the present invention will be apparent to those skilled in the art from consideration of the present specification and practice of the present invention disclosed herein. It is intended that the present specification and examples be considered as exemplary only with a true scope and spirit of the invention being indicated by the following claims and equivalents thereof.

Claims

1. A method for measuring edema, comprising: depressing a portion of skin of a patient at a location, by pressing a finger against skin of the patient, at the location, to apply pressure with the finger and cause a depression into the portion of the skin of the patient; releasing the pressure applied with the finger; casting a light pattern against the surface of the skin of the patient, including the depression, at the location, after the pressure is released, the light pattern being cast from a combined pattern-casting and imaging device and being distorted due to the depression; capturing an image of the distorted light pattern, with the combined pattern-casting and imaging device, within 15 seconds after the pressure is released; using a computing system to compare the image of the distorted light pattern with a preloaded pattern, to form a comparison; and using the computer system to calculate a depth of the depression, based on the comparison.
 2. The method of claim 1, wherein the pattern comprises a series of parallel lines.
 3. The method of claim 1, wherein the step of capturing an image of the distorted light pattern comprises processing image data of the captured image of the distorted light pattern.
 4. The method of claim 3, wherein the processing of the image data is carried out by the computing system, and the computing system is separate and apart from the combined pattern-casting and imaging device.
 5. The method of claim 3, wherein the computing system is part of the combined pattern-casting and imaging device and the combined pattern casting and imaging device is part of a smart device.
 6. The method of claim 3, further comprising wirelessly transmitting image data from the combined pattern casting and imaging device to the computing system, over a wireless network.
 7. The method of claim 6, wherein the image data is wirelessly transmitted as a live stream video, to the computing system, and the computing system is a remote computing system.
 8. The method of claim 1, wherein the combined pattern-casting and imaging device is a cell phone device comprising a flashlight feature, and the step of casting the light pattern against the surface of the skin of the patient comprises: fitting an adapter over the flashlight feature to form a pattern projector; and casting the light pattern with the pattern projector.
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